IoT-Based Smart Agriculture System for Environmental Monitoring and Crop Suggestion

1st Utkarsh Persai
23BCT0169
Vellore Institute of Technology
Vellore, India
Utkarsh.persai2023@vitstudent.ac.in

2nd Rishit Chaudhary
23BCT0157
Vellore Institute of Technology
Vellore, India
Rishit.chaudhary2023@vitstudent.ac.in

3rd Prof. Padma Priya R
Project Guide
Vellore Institute of Technology
Vellore, India

***Abstract*—Agriculture is the primary source of livelihood for a significant portion of the global population. However, the reliance on traditional, intuition-based farming practices often results in the inefficient utilization of natural resources, particularly fresh water. This project presents a comprehensive, basic Internet of Things (IoT) based smart agriculture system designed to optimize resource usage and reduce manual labor. Using an ESP32 microcontroller, a capacitive soil moisture sensor, and a DHT22 temperature and humidity sensor, the proposed architecture continuously monitors localized field conditions in real time. Agricultural telemetry is transmitted via serial communication to a local interface dashboard, which dynamically displays the current environmental state and provides a preliminary, rule-based crop suggestion. To further reduce human intervention, the system features a hardware-integrated automated irrigation mechanism utilizing a 5V relay and DC water pump, which activates autonomously when soil moisture saturates below a predefined analog threshold. This entry-level automated approach demonstrates a scalable methodology for enhancing yield potential and mitigating resource waste in small-to-medium scale farming operations.**

***Index Terms*—Internet of Things (IoT), Smart Agriculture, Precision Farming, ESP32, Sensor Networks, Automated Irrigation**

I. Introduction

Agriculture forms the backbone of the economy in many developing nations, contributing substantially to the Gross Domestic Product (GDP) and employing millions. Despite its immense economic importance, the sector is plagued by outdated methodologies. Farmers traditionally rely on historical weather patterns and manual inspections to determine when to irrigate and what crops to plant. This methodology is highly susceptible to human error, often leading to the over-watering of soil (causing root rot and water wastage) or under-watering (causing crop wilt). Furthermore, changing global climate patterns have made traditional intuition less reliable.

The advent of the Internet of Things (IoT) provides a highly cost-effective and scalable pathway to automate and digitize these agricultural processes. Precision agriculture leveraged through IoT allows for the micro-monitoring of soil and weather variables. In this project, an automated irrigation and monitoring system was conceptualized, designed, and developed to assist farmers in remote observation. The core objective is to collect real-time physical metrics—specifically ambient temperature, relative humidity, and volumetric soil moisture—and synthesize this data on a user-friendly graphical dashboard.

Beyond simple monitoring, the system implements a basic decision-support mechanism. By utilizing environmental thresholds, the dashboard can suggest the most biologically viable crop for current weather conditions, while the edge hardware autonomously dictates the mechanical flow of irrigation water, ensuring optimal soil saturation at all times.

II. Literature Review

The integration of electronics into agriculture has been a subject of extensive research over the past decade. Early implementations of automated irrigation largely relied on timer-based systems, which watered fields at specific intervals regardless of actual soil moisture, leading to severe inefficiencies during rainy seasons.

Recent advancements have shifted towards sensor-based IoT networks. For instance, systems utilizing the 8-bit Arduino Uno microcontroller established the viability of threshold-based moisture sensing, though they were limited by processing power and lack of native wireless capabilities. The transition to 32-bit hardware, such as the ESP8266 and ESP32 series, revolutionized field deployments by offering significant computational bandwidth alongside integrated Wi-Fi and Bluetooth stacks. While complex agricultural systems employ machine learning and drone-assisted visual imaging, there remains a critical need for low-cost, easily deployable base-level sensing architectures that can be maintained by farmers without deep technical expertise. This project seeks to fill that gap by focusing on high-reliability, low-complexity threshold automation.

III. Proposed System Architecture

The proposed architecture is compartmentalized into two distinct operational layers: the Hardware Sensing and Actuation Layer, and the Software Monitoring Layer.

A. Hardware Components

The physical node is centered around the ESP32 microcontroller, chosen for its fast clock speed, wide array of Analog-to-Digital Converter (ADC) pins, and robust power delivery.

- **DHT22 Environmental Sensor**: Unlike the standard DHT11, the DHT22 was selected for its superior operating range and precision. It measures ambient air temperature ($-40°$C to $80°$C) and relative humidity (0% to 100%) with high fidelity.
- **Capacitive Soil Moisture Sensor**: Inserted directly into the crop root zone, this sensor measures the volumetric water content. A capacitive sensor was specifically chosen over a resistive sensor because resistive probes suffer from rapid galvanic corrosion when exposed to damp soil and electrical current over time.
- **Mechanical Actuation (Relay and Pump)**: A standard 5V mechanical relay acts as an electronic switch. It isolates the high-current draw of the DC water pump from the delicate logic circuits of the ESP32, allowing the microcontroller to safely trigger irrigation sequences.

B. Software and Communication

The firmware written for the ESP32 utilizes a polling architecture, gathering analog reading averages over time to prevent mechanical "jittering" or false-triggers in the water pump due to slight electrical noise.

The collected data is packed and transmitted to a local computational interface via serial communication. A basic web dashboard serves two primary purposes: 1) **Real-Time Visualization**: It extracts the incoming telemetry stream and displays the current temperature, humidity, and soil moisture values on screen in human-readable formats. 2) **Rule-Based Crop Suggestion**: Using simple IF-ELSE conditional logic based on the sensor values, the software suggests a feasible crop category. For example, if the average ambient temperature is consistently high and the surrounding humidity is low, the system outputs a text recommendation for drought-resistant crops (e.g., Millet or Sorghum).

IV. Working Principle and Methodology

A. System Initialization and Calibration

Upon powering the device, the ESP32 undergoes a 5-second calibration phase where it establishes a baseline ambient reading. The capacitive moisture sensor continuously measures the dielectric permittivity of the surrounding soil, outputting an analog voltage inversely proportional to the soil wetness.

B. Autonomous Irrigation Logic

The primary objective of the hardware loop is preventing desiccation without human oversight. The ESP32 code contains a predefined critical moisture threshold (e.g., 30% relative saturation). If the rolling average of the analog reading falls below this threshold, indicating dry soil, the microcontroller asserts a digital HIGH signal to the relay module's control pin. This completes the secondary circuit, powering the immersed DC water pump. As water is dispersed over the soil, the capacitive sensor registers the rising moisture level. Once the threshold is safely exceeded ($> 60\%$ saturation), the ESP32 drops the pin to LOW, terminating the irrigation cycle. This hysteresis prevents the pump from rapidly turning on and off at the threshold line.

C. Dashboard User Interface

Concurrently, the ESP32 pushes formatted data strings over the USB-Serial connection at a baud rate of 115200. The connected local dashboard software parses these strings. The interface provides a persistent, at-a-glance view of farm conditions, rendering physical fieldwork largely unnecessary for simple daily checks.

V. Experimental Results and Observations

The prototype system was successfully assembled on a breadboard and subsequently tested in a controlled greenhouse environment.

A. Sensor Accuracy

During a continuous 48-hour testing period, the DHT22 provided stable temperature and humidity readings that correlated accurately with a benchmark commercial thermometer. The capacitive moisture sensor demonstrated high sensitivity, immediately registering the influx of water during the irrigation tests.

B. Actuation Reliability

The automated pump mechanism successfully operated without failure. When the plant soil was intentionally allowed to dry out, the relay triggered within 2 seconds of the moisture dropping past the critical threshold. No incidents of overwatering were recorded, proving the logic effectively managed the soil saturation without requiring constant manual oversight.

C. Software Operations

The serial data link suffered zero dropped packets during testing. The dashboard effectively displayed real-time telemetry, updating every 2 seconds. Furthermore, the rule-based crop suggestion engine accurately changed its output recommendations when the sensor node was moved between cold, damp environments and hot, dry simulated environments.

VI. Conclusion

This project successfully implemented a reliable, low-cost IoT-enabled agricultural monitoring network. By gathering granular data directly from the soil and local environment, the system effectively automated the cumbersome irrigation process utilizing localized threshold logic. The integration of a basic crop suggestion dashboard offers an accessible, easy-to-read interface for remote farm monitoring. The system fulfills its primary goal of reducing manual agricultural labor while ensuring optimal resource allocation.

VII. Future Work

While the current threshold-based system is functional, future iterations of this project could involve integrating broader wireless technologies such as LoRaWAN to communicate across much larger farmlands. Additionally, hardware could be expanded to include pH or NPK (Nitrogen, Phosphorus, Potassium) chemical sensors, allowing the rule-based dashboard to provide highly specific fertilizer recommendations alongside water management.

REFERENCES

[1] R. Nageswara Rao and B. Sridhar, "IoT based smart crop-field monitoring and automation irrigation system," *2018 2nd International Conference on Inventive Systems and Control (ICISC)*, 2018, pp. 478-481.

[2] K. Lakshmisudha, S. Hegde, N. Kale and S. Iyer, "Smart precision based agriculture using sensors," *International Journal of Computer Applications*, vol. 146, no. 11, pp. 36-38, 2016.

[3] S. R. Nandurkar, V. R. Thool and R. C. Thool, "Design and Development of Precision Agriculture System Using Wireless Sensor Network," *2014 First International Conference on Automation, Control, Energy and Systems (ACES)*, 2014, pp. 1-6.